

Week 4, Day 1 — student handout

Umesh Indrajith · SMS kickoff. No HTTP / Gin today.

TODAY'S OUTCOMES

- Package vs module; imports; internal/ boundary.
- go mod init and reading go.mod.
- Scaffold the SMS backend/ tree.
- go run ./cmd/server prints sms starting.
- README + PR. .env never committed.

SCAFFOLD

```
mkdir -p ~/epic-go/sms/backend && cd ~/epic-go/sms/backend
go mod init student-management-system
mkdir -p cmd/server
internal/{config,handlers,repository,models,routes,middleware,database,utils}
# add .gitkeep in each empty internal folder
```

CMD/SERVER/MAIN.GO

```
package main

import "fmt"

func main() {
    fmt.Println("sms starting")
}
```

```
gofmt -w cmd/server/main.go
go run ./cmd/server
```

TARGET TREE

```
backend/
  go.mod
  README.md
  .gitignore
  .env.example
cmd/server/main.go
internal/config/
internal/handlers/
internal/repository/
internal/models/
internal/routes/
internal/middleware/
```

```
internal/database/  
internal/utils/
```

WORDS

Term	Meaning
Module	Defined by go.mod; import prefix
Package	One folder, one package name
internal/	Compiler boundary for app code
cmd/	Executables / main packages

IF IT BREAKS

- not in std → create go.mod
- wrong folder → cd backend/
- empty dirs missing → .gitkeep
- .env in git → gitignore + rm --cached

HOMEWORK

Finish the skeleton as a **GitHub PR** before next class. No Gin yet. Details: homework.md.